

CARDIOVEX[®]

(cardiovexaban) tablets, for oral use

WARNING: BLEEDING RISK

CARDIOVEX increases the risk of bleeding and can cause significant and sometimes fatal bleeding. Do not use CARDIOVEX in patients with active pathological bleeding or a history of intracranial hemorrhage.

Most bleeding events with CARDIOVEX are gastrointestinal. Intracranial hemorrhage rates are low and comparable to placebo (0.4% vs 0.3%). Monitor patients for signs and symptoms of bleeding. Discontinue CARDIOVEX in patients with active bleeding.

1 INDICATIONS AND USAGE

CARDIOVEX is a dual P2Y₁₂/PAR-1 platelet inhibitor indicated to reduce the risk of thrombotic cardiovascular events in adult patients with established atherosclerotic cardiovascular disease (ASCVD) and elevated residual risk despite guideline-directed medical therapy.

Specific Patient Population:

- Documented ASCVD: Prior myocardial infarction (>30 days), prior ischemic stroke (>90 days), or symptomatic peripheral arterial disease with objective evidence
- Elevated residual risk: 10-year ASCVD risk score $\geq 15\%$ OR ≥ 2 high-risk features (diabetes with target organ damage, chronic kidney disease stage 3+, or recurrent events within 24 months)
- On guideline-directed medical therapy: Stable antiplatelet (aspirin or clopidogrel), high-intensity statin with LDL <100 mg/dL, and blood pressure control

2 DOSAGE AND ADMINISTRATION

2.1 Recommended Dosage

The recommended dosage of CARDIOVEX is 150 mg orally once daily with or without food.

Continue background antiplatelet therapy (aspirin 75-100 mg daily OR clopidogrel 75 mg daily, not both) and statin therapy.

2.2 Switching from Other Antiplatelet Agents

When switching from clopidogrel or ticagrelor, initiate CARDIOVEX 24 hours after the last dose. When switching from prasugrel, initiate CARDIOVEX 48 hours after the last dose.

3 DOSAGE FORMS AND STRENGTHS

CARDIOVEX tablets: 150 mg, round, blue, film-coated tablets debossed with 'CVX 150' on one side

4 CONTRAINDICATIONS

CARDIOVEX is contraindicated in patients with:

- Active pathological bleeding (e.g., peptic ulcer, intracranial hemorrhage)

- History of intracranial hemorrhage
- Severe hepatic impairment (Child-Pugh C)
- Hypersensitivity to cardiovexaban or any component of CARDIOVEX

5 WARNINGS AND PRECAUTIONS

5.1 Increased Risk of Bleeding

CARDIOVEX increases the risk of bleeding. In the SHIELD-1 trial, TIMI major bleeding occurred in 3.8% of CARDIOVEX-treated patients versus 2.1% with placebo (HR 1.82, 95% CI 1.39-2.38, $p<0.001$) over a median of 36 months. Most bleeding events (70%) were gastrointestinal.

Intracranial hemorrhage rates were low and balanced: 0.4% (CARDIOVEX) versus 0.3% (placebo), HR 1.23, 95% CI 0.59-2.58, $p=0.58$. This finding was confirmed in SHIELD-2 (0.4% vs 0.3%, HR 1.36, $p=0.44$) and the pooled analysis of 16,048 patients (HR 1.29, $p=0.35$).

5.2 Discontinuation for Surgery and Invasive Procedures

Discontinue CARDIOVEX 5-7 days prior to elective surgery or invasive procedures with a major risk of bleeding. Bridging antiplatelet therapy is generally not recommended. Resume CARDIOVEX when hemostasis has been established.

5.3 Concomitant Use with Other Antiplatelet Agents

CARDIOVEX was studied with background aspirin OR clopidogrel monotherapy. Concomitant use with dual antiplatelet therapy (aspirin plus P2Y₁₂ inhibitor) has not been studied and increases bleeding risk. Do not use CARDIOVEX with dual antiplatelet therapy beyond 12 months post-ACS or post-PCI.

5.4 Gastrointestinal Bleeding Risk Factors

Patients with prior gastrointestinal bleeding, active peptic ulcer disease, inflammatory bowel disease, or concurrent use of medications that increase bleeding risk (anticoagulants, chronic NSAIDs, SSRIs/SNRIs, corticosteroids) are at increased risk for GI bleeding. Consider proton pump inhibitor therapy in patients at high risk.

6 ADVERSE REACTIONS

6.1 Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

The safety of CARDIOVEX was evaluated in SHIELD-1 (N=4,103) and SHIELD-2 (N=3,921), randomized placebo-controlled trials in patients with established ASCVD on guideline-directed therapy. The median duration of exposure was 36 months.

Table 1. Bleeding Events in SHIELD Trials (Pooled Analysis, N=16,048)

Bleeding Event	CARDIOVEX (N=8024)	Placebo (N=8024)	Hazard Ratio (95% CI)	p-value
TIMI Major Bleeding	316 (3.9%)	176 (2.2%)	1.80 (1.48-2.18)	<0.001
Intracranial hemorrhage	31 (0.4%)	24 (0.3%)	1.29 (0.76-2.20)	0.35
Gastrointestinal bleeding	221 (2.8%)	119 (1.5%)	1.86 (1.48-2.34)	<0.001
Fatal bleeding	23 (0.3%)	15 (0.2%)	1.53 (0.80-2.94)	0.20
TIMI Minor Bleeding	580 (7.2%)	340 (4.2%)	1.71 (1.49-1.97)	<0.001

6.2 Non-Bleeding Adverse Reactions

Non-bleeding adverse reactions reported in $\geq 2\%$ of CARDIOVEX-treated patients and more frequently than placebo: dyspepsia (3.2% vs 1.8%), nausea (2.7% vs 1.4%), headache (2.3% vs 1.9%).

7 DRUG INTERACTIONS

7.1 Strong CYP3A Inhibitors

Avoid concomitant use of CARDIOVEX with strong CYP3A inhibitors (e.g., ketoconazole, itraconazole, clarithromycin, ritonavir). These agents increase cardiovexaban exposure and bleeding risk.

7.2 Strong CYP3A Inducers

Avoid concomitant use with strong CYP3A inducers (e.g., rifampin, carbamazepine, phenytoin, St. John's wort). These agents may reduce cardiovexaban exposure and efficacy.

7.3 Anticoagulants and Antiplatelet Agents

Concomitant use with oral anticoagulants, direct thrombin inhibitors, or Factor Xa inhibitors increases bleeding risk. Avoid concomitant use unless benefit outweighs risk. CARDIOVEX was studied with background single antiplatelet therapy (aspirin OR clopidogrel), not dual antiplatelet therapy.

8 USE IN SPECIFIC POPULATIONS

8.1 Pregnancy

Risk Summary: There are no adequate and well-controlled studies of CARDIOVEX in pregnant women. Animal reproduction studies have not been conducted. It is not known whether CARDIOVEX can cause fetal harm when administered to a pregnant woman.

8.2 Lactation

Risk Summary: It is not known whether cardiovexaban is present in human milk. Because many drugs are present in human milk and because of the potential for serious adverse reactions in nursing infants, a decision should be made whether to discontinue nursing or to discontinue the drug.

8.4 Pediatric Use

Safety and effectiveness in pediatric patients have not been established.

8.5 Geriatric Use

Of the 8,024 patients in SHIELD-1 and SHIELD-2 who received CARDIOVEX, 3,850 (48%) were ≥ 65 years and 1,284 (16%) were ≥ 75 years. No overall differences in safety or effectiveness were observed between these patients and younger patients.

8.6 Renal Impairment

Mild to moderate renal impairment (eGFR 30-89 mL/min/1.73 m²) does not require dose adjustment. CARDIOVEX has not been studied in patients with severe renal impairment (eGFR <30 mL/min/1.73 m²) or on dialysis. Use with caution in these populations.

8.7 Hepatic Impairment

CARDIOVEX is contraindicated in patients with severe hepatic impairment (Child-Pugh C). No dosage adjustment is needed for patients with mild or moderate hepatic impairment (Child-Pugh A or B).

10 OVERDOSAGE

There is no known specific antidote to CARDIOVEX. Treatment of overdose should follow general supportive measures. Monitor for signs of bleeding and manage bleeding events with standard supportive care including transfusion support as needed. Due to the antiplatelet mechanism, dialysis is unlikely to be effective. Cardiovoxaban's antiplatelet effects dissipate within 3-5 days of drug discontinuation based on pharmacodynamic studies.

11 DESCRIPTION

CARDIOVEX (cardiovoxaban) is a dual-pathway antiplatelet agent that selectively and reversibly inhibits both the P2Y₁₂ receptor (blocking ADP-mediated platelet activation) and the protease-activated receptor-1 (PAR-1, blocking thrombin-mediated platelet activation).

The chemical name is (synthetic structure not disclosed for fictional compound). The molecular formula is C₂₄H₂₇N₅O₃S with a molecular weight of 465.57 g/mol. Cardiovoxaban is a white to off-white crystalline powder, slightly soluble in water.

Each CARDIOVEX tablet contains 150 mg of cardiovoxaban and the following inactive ingredients: microcrystalline cellulose, croscarmellose sodium, hypromellose, magnesium stearate, titanium dioxide, polyethylene glycol, and FD&C; Blue #2 aluminum lake.

12 CLINICAL PHARMACOLOGY

12.1 Mechanism of Action

Cardiovoxaban and its active metabolite inhibit platelet aggregation through dual pathways. P2Y₁₂ receptor antagonism blocks ADP-induced platelet activation and aggregation. PAR-1 receptor antagonism blocks thrombin-induced platelet activation. Dual inhibition provides more complete suppression of platelet activation than single-pathway strategies.

12.2 Pharmacodynamics

In phase 2 studies, cardiovoxaban 150 mg once daily demonstrated superior platelet inhibition compared with clopidogrel 75 mg daily as measured by ADP-induced and thrombin-induced platelet aggregation assays. Maximal platelet inhibition is achieved within 2 hours of dosing. Pharmacodynamic effects are sustained over 24 hours, consistent with once-daily dosing.

12.3 Pharmacokinetics

Absorption: Cardiovoxaban is rapidly absorbed after oral administration with peak plasma concentrations (C_{max}) occurring at 1-2 hours post-dose. Absolute bioavailability is approximately 75%. Food does not significantly affect exposure.

Distribution: Cardiovoxaban is highly protein-bound (>95%) to albumin and alpha-1-acid glycoprotein. Volume of distribution at steady state is approximately 165 L.

Metabolism: Cardiovoxaban is primarily metabolized by CYP3A4 to an active metabolite. Both parent compound and metabolite contribute to pharmacological activity.

Excretion: Cardiovoxaban is eliminated primarily via hepatic metabolism and biliary excretion. Terminal half-life is approximately 8-10 hours, supporting once-daily dosing.

14 CLINICAL STUDIES

14.1 Secondary Prevention in High-Risk ASCVD (SHIELD-1 and SHIELD-2)

The efficacy of CARDIOVEX was established in two randomized, double-blind, placebo-controlled trials: SHIELD-1 (N=8,206) and SHIELD-2 (N=7,842). Both trials enrolled patients with established ASCVD (prior MI, prior ischemic stroke, or symptomatic PAD) and elevated residual risk (ASCVD risk score $\geq 15\%$ or ≥ 2 high-risk features) despite guideline-directed medical therapy including antiplatelet agent (aspirin or clopidogrel) and high-intensity statin.

The primary endpoint was time to first MACCE (cardiovascular death, myocardial infarction, stroke, or urgent coronary revascularization). Median follow-up was 36 months in both trials.

Table 2. Efficacy Results in SHIELD-1

Endpoint	CARDIOVEX (N=4103)	Placebo (N=4103)	Hazard Ratio (95% CI)	p-value
MACCE (primary)	489 (11.9%)	643 (15.7%)	0.75 (0.64-0.88)	<0.001
3-point MACE	412 (10.0%)	531 (12.9%)	0.77 (0.65-0.91)	0.002
CV death	142 (3.5%)	181 (4.4%)	0.78 (0.61-1.01)	0.06
MI	198 (4.8%)	268 (6.5%)	0.73 (0.59-0.90)	0.003
Stroke	108 (2.6%)	157 (3.8%)	0.68 (0.52-0.89)	0.005

SHIELD-2 confirmed the findings of SHIELD-1. The primary endpoint (MACCE) occurred in 11.5% of CARDIOVEX patients versus 15.1% of placebo patients (HR 0.73, 95% CI 0.63-0.85, $p < 0.001$). The stroke reduction was consistent with SHIELD-1 (HR 0.68, 95% CI 0.51-0.89, $p = 0.006$).

In the pooled analysis of 16,048 patients, CARDIOVEX reduced MACCE by 26% (HR 0.74, 95% CI 0.66-0.82, $p < 0.001$) and stroke by 32% (HR 0.68, 95% CI 0.56-0.83, $p < 0.001$). Cardiovascular death was reduced by 22% (HR 0.78, 95% CI 0.65-0.94, $p = 0.009$).

16 HOW SUPPLIED/STORAGE AND HANDLING

16.1 How Supplied

CARDIOVEX (cardiovexaban) tablets are supplied as:

150 mg tablets: Round, blue, film-coated tablets debossed with 'CVX 150' on one side

- Bottles of 30: NDC 12345-678-30
- Bottles of 90: NDC 12345-678-90

16.2 Storage

Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Keep tablets in original container. Protect from moisture.

17 PATIENT COUNSELING INFORMATION

Advise the patient to read the FDA-approved patient labeling (Medication Guide).

Bleeding Risk

Inform patients that CARDIOVEX increases the risk of bleeding and to report any unanticipated, prolonged, or excessive bleeding, or blood in stool or urine. Advise patients to inform physicians and dentists that they are taking CARDIOVEX before any surgery or dental procedure is scheduled and before any new drug is taken.

Adherence to Therapy

Instruct patients to take CARDIOVEX exactly as prescribed and not to discontinue without first discussing with their healthcare provider. Inform patients that premature discontinuation increases the risk of cardiovascular events.

Missed Doses

If a dose is missed, instruct patients to take CARDIOVEX at the next scheduled time and not to double the dose.

DISCLAIMER

This is a fictional FDA label created for educational and demonstration purposes only. CARDIOVEX (cardiovexaban) is not a real drug. Novellia Therapeutics is a fictional entity. All clinical trial data (SHIELD-1, SHIELD-2) are simulated. This document should not be used for clinical decision-making, regulatory submission, or any purpose other than training and education.

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